PCBC H Name _____ _____ /22

Limits Quiz 1.1–1.2

Non-Calculator

You have 20 minutes.

PRACTICE A · built to match the real Quiz 1.1–1.2: 11 questions, 20 minutes, no calculator

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.

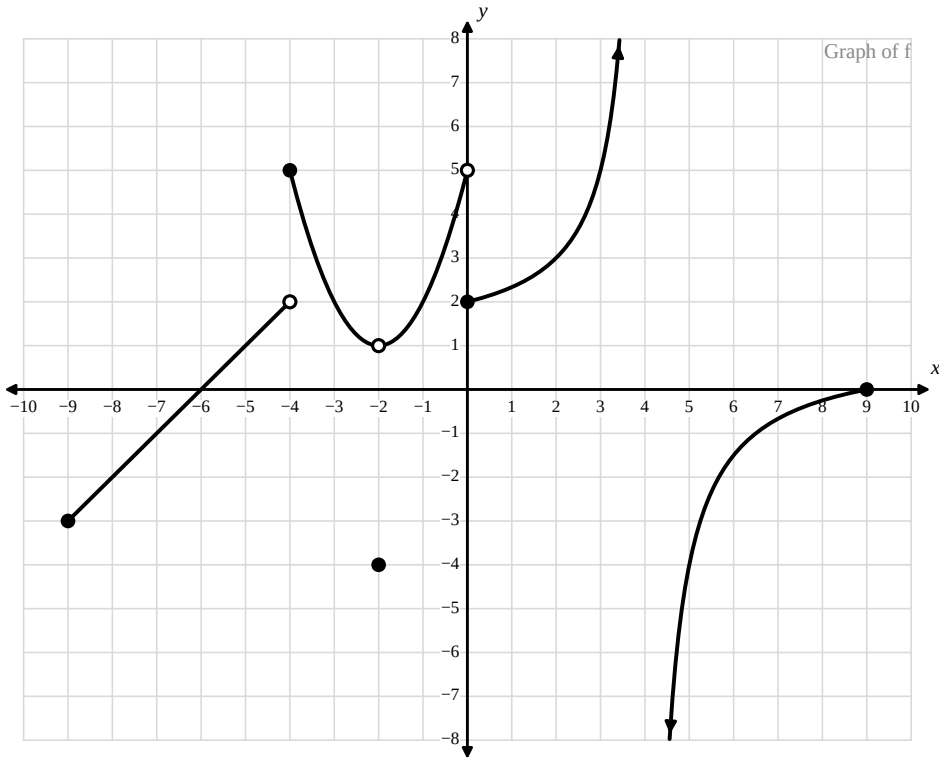
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Signature _____

SHOW ALL WORK FOR FULL CREDIT.

Use the graph of f shown below to find each. If a limit does not exist, write DNE. (2 points each)

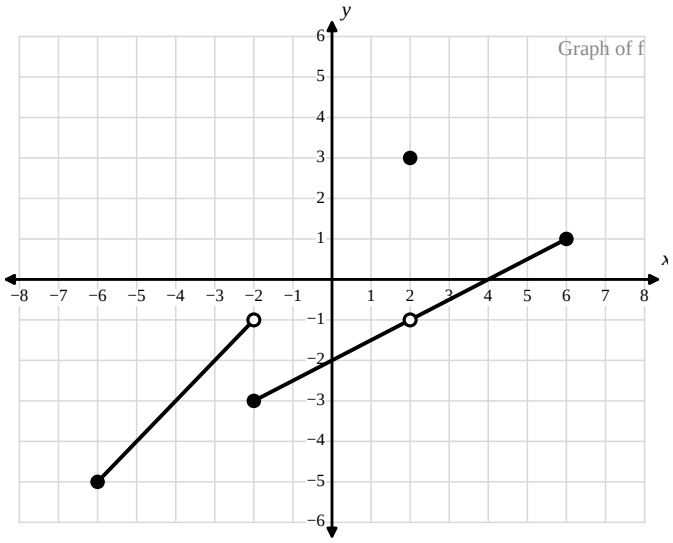


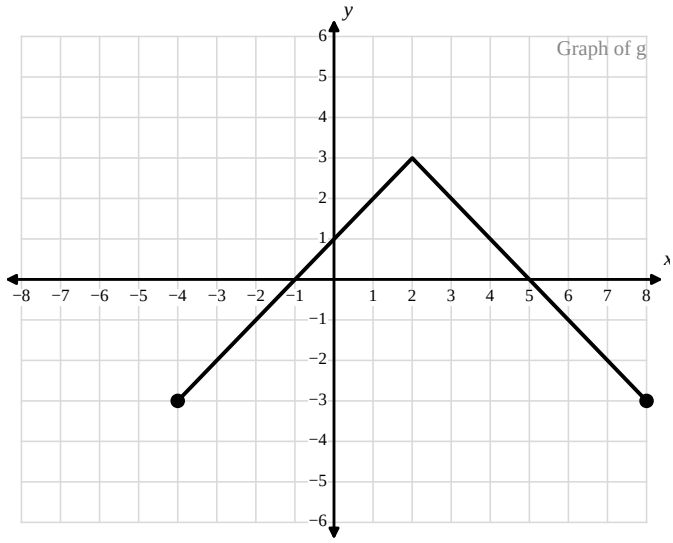
1. $\lim_{x\to -2} f(x) =$	**2.** $\lim_{x\to -4} f(x) =$	**3.** $\lim_{x\to 4^+} f(x) =$

(2 points)

4. Use the graph of f above to determine all values of a for which $\lim_{x\to a} f(x) = 2$ on the interval $-9 < x < 9$. If there are no values of a, write the empty set.

Use the graphs of f and g below to find the limit. Show the limit properties you use. (2 points)





5. $\lim_{x \to -2^+} [3f(x) - g(x)]$

(2 points)

6. Use the table to estimate $\lim_{x \to 3} g(x)$. Give the left-hand limit, the right-hand limit, and the limit, or state that it does not exist.

x	2.9	2.99	2.999	$\to 3 \leftarrow$	3.001	3.01	3.1
$g(x)$	0.2515823	0.2501564	0.2500156		0.2499844	0.2498439	0.2484567

$\lim_{x \to 3^-} g(x) \approx$ _____ $\lim_{x \to 3^+} g(x) \approx$ _____ $\lim_{x \to 3} g(x) \approx$ _____

Evaluate each limit. (2 points each)

7. $\lim\limits_{x \to 3^-} \dfrac{x-3}{|x-3|}$

8. $\lim\limits_{x \to -3} \dfrac{x^2+x-6}{x^2-9}$

9. $\lim\limits_{x \to 2} \dfrac{x-2}{\sqrt{2x+5}-3}$

10. $\lim\limits_{x \to 0} \dfrac{\frac{1}{x+3} - \frac{1}{3}}{x}$

11. For $f(x) = x^2 - 5x + 2$, find $\lim\limits_{h \to 0} \dfrac{f(x+h) - f(x)}{h}$

PCBC H Name **ANSWER KEY** **22** /22

Limits Quiz 1.1–1.2

Non-Calculator

You have 20 minutes.

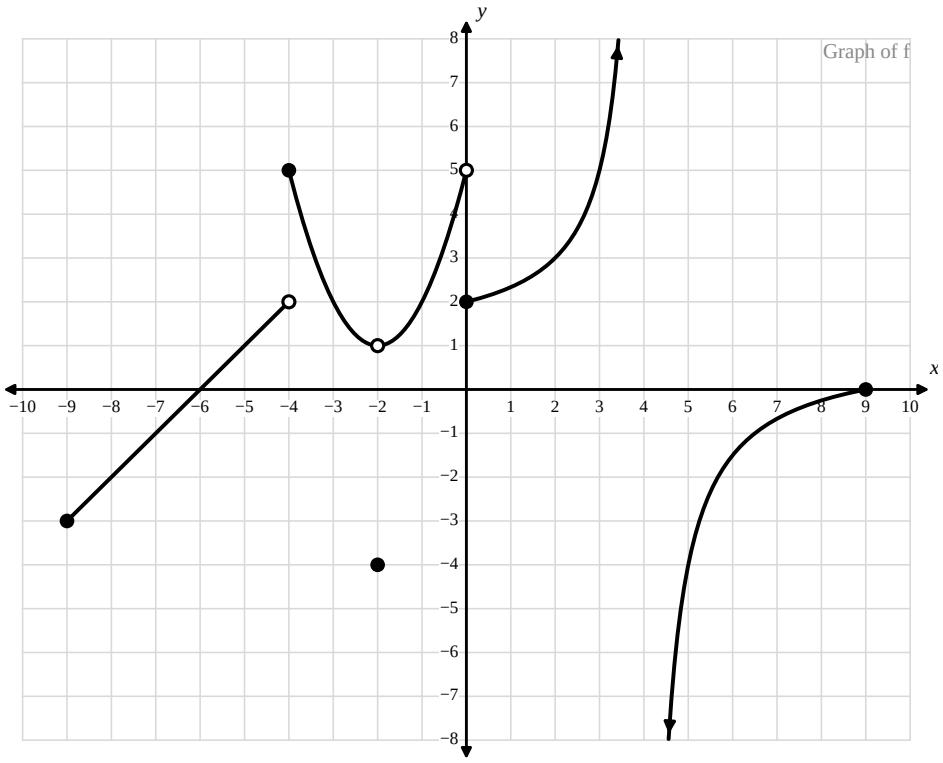
PRACTICE A · built to match the real Quiz 1.1–1.2: 11 questions, 20 minutes, no calculator

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SHOW ALL WORK FOR FULL CREDIT.

Use the graph of f shown below to find each. If a limit does not exist, write DNE. (2 points each)



1. $\lim_{x\to -2} f(x) =$

Both sides of the curve head to the open circle at $(-2, 1)$.

1

2. $\lim_{x\to -4} f(x) =$

$\lim_{x\to -4^-} f(x) = 2$, $\lim_{x\to -4^+} f(x) = 5$

$2 \neq 5$, *so* $\lim_{x\to -4} f(x)$ *does not exist.*

DNE

3. $\lim_{x\to 4^+} f(x) =$

Just right of $x = 4$ *the curve falls without bound.*

$-\infty$

(2 points)

4. Use the graph of f above to determine all values of a for which $\lim_{x\to a} f(x) = 2$ on the interval $-9 < x < 9$. If there are no values of a, write the empty set.

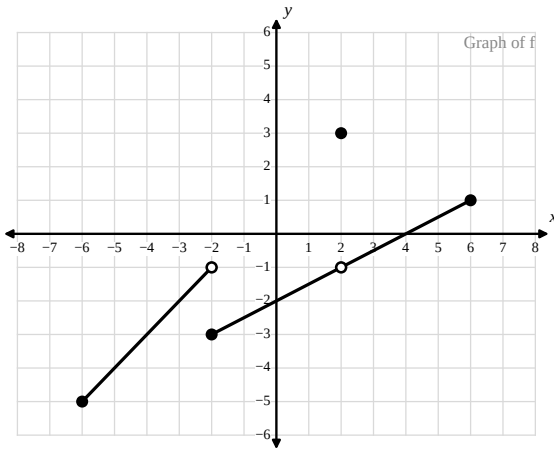
The curve reaches height 2 from both sides at $x = -3$ *and* $x = -1$.

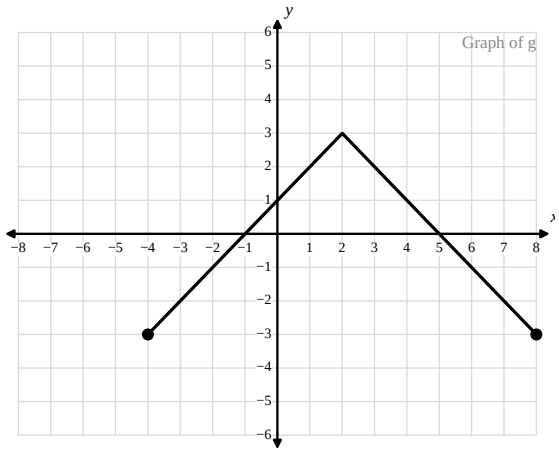
Not $x = -4$*: only the left side heads to 2 (the right side starts at 5).*

Not $x = 0$*: only the right side starts at 2 (the left side heads to 5).*

$a = -3,\ a = -1$

Use the graphs of f and g below to find the limit. Show the limit properties you use. (2 points)





5. $\lim_{x \to -2^+} [3f(x) - g(x)]$

$\lim_{x \to -2^+} [3f(x) - g(x)] = 3 \lim_{x \to -2^+} f(x) - \lim_{x \to -2^+} g(x)$

$= 3(-3) - (-1)$

$= -8$

-8

(2 points)

6. Use the table to estimate $\lim_{x \to 3} g(x)$. Give the left-hand limit, the right-hand limit, and the limit, or state that it does not exist.

x	2.9	2.99	2.999	$\to 3 \leftarrow$	3.001	3.01	3.1
$g(x)$	0.2515823	0.2501564	0.2500156		0.2499844	0.2498439	0.2484567

Left: 0.2515823, 0.2501564, 0.2500156

Right: 0.2484567, 0.2498439, 0.2499844

Both sides agree. A table estimates; it does not prove.

≈ 0.25

$\lim_{x \to 3^-} g(x) \approx$ 0.25 $\lim_{x \to 3^+} g(x) \approx$ 0.25 $\lim_{x \to 3} g(x) \approx$ 0.25

Evaluate each limit. (2 points each)

7. $\lim\limits_{x\to 3^-} \dfrac{x-3}{|x-3|}$

$$\lim_{x\to 3^-} \frac{x-3}{|x-3|} = \lim_{x\to 3^-} \frac{x-3}{-(x-3)}$$
$$= \lim_{x\to 3^-} (-1) = -1$$

-1

8. $\lim\limits_{x\to -3} \dfrac{x^2+x-6}{x^2-9}$

Substitution produces 0/0: indeterminate.

$$\lim_{x\to -3} \frac{x^2+x-6}{x^2-9} = \lim_{x\to -3} \frac{(x+3)(x-2)}{(x+3)(x-3)}$$
$$= \lim_{x\to -3} \frac{x-2}{x-3}$$
$$= \frac{-3-2}{-3-3} = \frac{-5}{-6} = \frac{5}{6}$$

$\frac{5}{6}$

9. $\lim\limits_{x\to 2} \dfrac{x-2}{\sqrt{2x+5}-3}$

Substitution produces 0/0: indeterminate.

$$\lim_{x\to 2} \frac{x-2}{\sqrt{2x+5}-3} = \lim_{x\to 2} \frac{(x-2)\left(\sqrt{2x+5}+3\right)}{(2x+5)-9}$$
$$= \lim_{x\to 2} \frac{(x-2)\left(\sqrt{2x+5}+3\right)}{2(x-2)}$$
$$= \lim_{x\to 2} \frac{\sqrt{2x+5}+3}{2}$$
$$= \frac{\sqrt{9}+3}{2} = \frac{6}{2} = 3$$

3

10. $\lim\limits_{x\to 0} \dfrac{\frac{1}{x+3} - \frac{1}{3}}{x}$

Substitution produces 0/0: indeterminate.

$$\lim_{x\to 0} \frac{\frac{1}{x+3} - \frac{1}{3}}{x} = \lim_{x\to 0} \frac{3-(x+3)}{3x(x+3)}$$
$$= \lim_{x\to 0} \frac{-x}{3x(x+3)}$$
$$= \lim_{x\to 0} \frac{-1}{3(x+3)}$$
$$= \frac{-1}{3(0+3)} = -\frac{1}{9}$$

$-\frac{1}{9}$

11. For $f(x) = x^2 - 5x + 2$, find $\lim\limits_{h\to 0} \dfrac{f(x+h)-f(x)}{h}$

$$\lim_{h\to 0} \frac{\left[(x+h)^2 - 5(x+h) + 2\right] - \left(x^2 - 5x + 2\right)}{h}$$
$$= \lim_{h\to 0} \frac{x^2 + 2xh + h^2 - 5x - 5h + 2 - x^2 + 5x - 2}{h}$$
$$= \lim_{h\to 0} \frac{h(2x + h - 5)}{h}$$
$$= \lim_{h\to 0} (2x + h - 5) = 2x + 0 - 5 = 2x - 5$$

$2x - 5$